HMMamba: Hierarchical Multiscale Mamba Network for Joint Classification of Hyperspectral and LiDAR Data

Cuiping Shi, *Member, IEEE*, Yeqi Lei, Fei Zhu, Diling Liao, *Student Member, IEEE*, Chenyang Fu, and Liguo Wang, *Member, IEEE*

***Abstract*—The joint classification of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data is an important research direction in the field of remote sensing image processing. However, existing methods are mostly limited to inadequate feature extraction or simple serial fusion strategies, failing to fully exploit the deep cross-modal relationships between LiDAR and HSI data. Meanwhile, global modeling based on transformers, despite capturing long-range dependencies, faces difficulties in handling high-dimensional remote sensing data due to quadratic computational complexity, leading to constrained global–local feature collaboration. To address these issues, this article proposes a novel hierarchical multiscale Mamba (HMMamba) network for joint classification of HSI and LiDAR data. First, the method adopts a hierarchical design to obtain rich multilevel feature representations through multiscale feature extraction. Second, it utilizes adaptive weight allocation to achieve dynamic cross-modal feature integration. Finally, it constructs a fused feature enhancement Mamba block (FFMB), which integrates a dual-branch attention mechanism and selective state space modeling to achieve efficient global context modeling under linear computational complexity. Extensive experiments on four public datasets (Houston2013, MUUFL, Trento, and Augsburg) demonstrate that HMMamba significantly outperforms existing state-of-the-art methods in terms of classification performance. Specifically, on the Augsburg dataset, the overall accuracy (OA) improvement is 3.24% compared to the suboptimal method, fully validating the superior performance of the proposed method. The code link of the proposed method is https://github.com/leiyeqi/HMMamba**

***Index Terms*—Classification, hyperspectral image (HSI), light detection and ranging (LiDAR), Mamba, multimodal fusion.**

Received 10 October 2025; accepted 15 November 2025. Date of publication 18 November 2025; date of current version 1 December 2025. This work was supported in part by the Science and Technology Plan Project of Huzhou under Grant 2024GZ36, in part by the Open Funding of China-Mozambique "Belt and Road" Joint Laboratory on Smart Agriculture under Grant CMSA-KF202510, in part by the National Natural Science Foundation of China 42271409, and in part by the National College Student Innovation Training Program Project under Grant 202510347021. *(Corresponding author: Cuiping Shi.)*

Cuiping Shi is with the College of Information Engineering, Huzhou University, Huzhou 313000, China, also with the China-Mozambique "Belt and Road" Joint Laboratory on Smart Agriculture, Zhejiang Normal University, Jinhua 321004, China, and also with the Huzhou Key Laboratory of Waters Robotics Technology, Huzhou University, Huzhou 313000, China (e-mail: shicuiping@zjhu.edu.cn).

Yeqi Lei is with the College of Information Engineering, Huzhou University, Huzhou 313000, China (e-mail: 2023388418@stu.zjhu.edu.cn).

Fei Zhu is with the School of Geospatial Artificial Intelligence, East China Normal University, Shanghai 200241, China (e-mail: 52290161013@stu.ecnu.edu.cn).

Diling Liao is with the School of Electronics and Communication Engineering, Sun Yat-sen University, Shenzhen 518000, China (e-mail: liaodling@mail2.sysu.edu.cn).

Chenyang Fu is with the Anding Academy, Huzhou University, Huzhou 313000, China (e-mail: 2024082201@stu.zjhu.edu.cn).

Liguo Wang is with the College of Information and Communication Engineering, Dalian Nationalities University, Dalian 116000, China (e-mail: wangliguo@hrbeu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3634369

I. Introduction

With the rapid advancement of remote sensing technology, multimodal data collaborative processing has emerged as a core research direction in the field of Earth observation. Hyperspectral image (HSI) captures fine spectral signatures of ground objects through hundreds of continuous narrow bands, playing a crucial role in land cover identification and environmental monitoring [1], [2]. However, single-modal data exhibits limitations in complex scene analysis [3]. Light detection and ranging (LiDAR) acquires precise 3-D geometric information through active laser ranging, forming excellent complementarity with the spectral information of HSI [4], [5]. Consequently, joint classification of HSI and LiDAR data has become an important research hotspot in the field of remote sensing image processing [6].

In recent years, deep learning-based multimodal fusion methods have achieved significant progress. Convolutional neural networks (CNNs) have been widely adopted due to their powerful spatial feature extraction capability [7], [8], but are limited by restricted receptive fields, making it challenging to capture long-range dependencies [9]. The transformer architecture enhances global modeling capability through self-attention mechanisms [10], [11], yet its $O(N^2)$ computational complexity constrains its application on large-scale data. In the domain of HSI-LiDAR joint classification, notable advances have been made by methods such as GLT-Net proposed by Ding et al. [12], HCT designed by Zhao et al. [13], and ExViT introduced by Yao et al. [14]. Recently, DSHFNet designed by Feng et al. [15], CrossHL developed by Roy et al. [16], S^2EFT proposed by Feng et al. [17], and the hierarchical multimodal feature aggregation method based on multihead axial attention transformer proposed by Zhu et al. [18] have further advanced this field.

It is noteworthy that developments in deep learning technology have demonstrated tremendous potential not only in HSI-LiDAR joint classification but also in other multimodal learning domains. For instance, FSVLM and FarmSeg_VLM proposed by Wu et al. [19], [20] achieved high-precision farmland remote sensing segmentation through vision-language

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

alignment techniques, providing important references for multimodal fusion. Graph neural networks have also shown advantages in processing non-Euclidean data, with GAMF proposed by Cai et al. [21] exploring the correlation and complementarity of HSI and LiDAR data.

Algorithm 1 Process of Mamba Sequence Modeling

Input: Fused multi-scale feature $F^{\text{fused}} \in \mathbb{R}^{N_b \times N \times C}$ from multi-scale integration

Normalize: $\hat{F}^{\text{fused}} = \text{LayerNorm}(F^{fused})$
// Dual-Branch Projection
Project to SSM branch: $X_{\text{ssm}} = \text{Linear}_{\text{ssm}}(\hat{F}^{\text{fused}}) \in \mathbb{R}^{N_b \times N \times D}$;
Project to gate branch: $X_{\text{gate}} = \text{Linear}_{\text{gate}}(\hat{F}^{\text{fused}}) \in \mathbb{R}^{N_b \times N \times D}$, where $D = 2C$;
//Selective State Space Evolution
Initialize hidden state: $h_0 = 0$;
for $t = 1$ **to** N **do**
    Extract input: $x_t = X_{\text{ssm}}[:, t, :]$;
    //Content-dependent parameter generation
    Generate time step: $\Delta_t = \text{softplus}(\text{Linear}_{\Delta}(x_t))$
    Generate input gate: $B_t = \text{Linear}_B(x_t)$
    Generate output gate: $C_t = \text{Linear}_C(x_t)$
    //State evolution with selective mechanism
    Update state: $h_t = \exp(\Delta_t \cdot A) \odot h_{t-1} + (\Delta_t \cdot B_t) \odot x_t$
    //Output projection
    Compute output: $y_t = C_t \odot h_t$
//Gated Fusion and Residual Connection
Collect sequence outputs: $Y_{\text{ssm}} = [y_1, y_2, \ldots, y_N]$
Apply gated fusion: $Z = X_{\text{gate}} \odot \text{SiLU}(Y_{\text{ssm}})$
Add residual connection: $F^{\text{out}} = \text{Linear}_{\text{proj}}(Z) + F^{\text{fused}}$

Output: Enhanced global feature $F^{\text{out}} \in \mathbb{R}^{N_b \times N \times C}$

Recently, state space models (SSMs), particularly the Mamba architecture, have provided a new paradigm for sequence modeling [22], [23]. Compared to transformers, Mamba achieves effective long-sequence modeling while maintaining $O(N)$ linear complexity. In the remote sensing domain, ChangeMamba proposed by Chen et al. [24] has been applied to change detection, and RS3Mamba developed by Ma et al. [25] for semantic segmentation. MSFMamba proposed by Gao et al. [26] achieved multiscale feature fusion through hierarchical state space modeling. Liao et al. [27] first introduced Mamba into HSI-LiDAR joint classification, proposing the HLMamba method. Shi et al. [28] further developed TBi-Mamba, employing bidirectional Mamba blocks to capture forward and backward dependencies.

Beyond supervised learning, self-supervised learning has provided new pathways for scenarios with limited labeled data. S^2MAE proposed by Lin et al. [29] learns robust feature representations through masked prediction, NNCL introduced by Wang et al. [30] constructs contrastive pairs using nearest neighbor samples, and PICN developed by Gao et al. [31] compensates for information loss through prototype learning. In terms of multiscale feature fusion, FPN proposed by Lin et al. [32] and DenseNet designed by Huang et al. [33] have provided fundamental frameworks for multiscale modeling. Attention mechanisms play a critical role in feature fusion, with CBAM proposed by Woo et al. [34] and the SE module designed by Hu et al. [35] demonstrating the effectiveness of channel and spatial attention.

Despite significant progress in HSI-LiDAR joint classification, existing research still faces three key challenges: 1) cross-modal fusion strategies predominantly employ fixed weights or simple concatenation, unable to dynamically adjust modal weights according to different land cover types (e.g., vegetation relying on spectra and buildings relying on elevation); 2) although multiscale extraction is adopted, the lack of interscale interaction mechanisms prevents modeling of dependencies and complementarity across different observation levels; and 3) fused features require simultaneous modeling of channel importance and spatial saliency while capturing global dependencies, where the local receptive field of CNNs and the quadratic complexity of transformers struggle to balance efficiency and capability.

To address these challenges, this article proposes a hierarchical multiscale Mamba (HMMamba) network for HSI-LiDAR joint classification. The main contributions include as follows.

1) An adaptive weight fusion (AWF) module is designed, which can achieve scene-adaptive modal weight allocation through the synergy of global learnable parameters and local gating networks.
2) A multiscale attention fusion (MSAF) module is constructed, which can automatically identify optimal observation scales for different land cover types through competitive attention mechanisms.
3) A fused feature enhancement Mamba block (FFMB) is proposed. It can combine dual-branch attention with selective SSMs to achieve refined processing and global dependency modeling of fused features under $O(N)$ linear complexity. Experiments on four publicly available datasets demonstrate that HMMamba significantly outperforms existing methods in both classification accuracy and computational efficiency.

The remainder of this study is organized as follows. In Section II, related work on HSI-LiDAR joint classification is systematically reviewed, and the advantages and limitations of existing methods are analyzed. In Section III, the architectural design of the HMMamba network and the technical implementation details of each core module are elaborated. In Section IV, a comprehensive experimental analysis is conducted, including experimental setup, quantitative comparisons, ablation studies, and visualization analysis to validate the proposed method. In Section V, conclusions are drawn and future research directions are discussed.

II. Related Work

Research on HSI and LiDAR joint classification has undergone an evolutionary process from traditional methods to deep learning approaches. In recent years, deep learning, particularly CNNs, has brought new opportunities for multimodal remote sensing data fusion.

A. Multimodal Fusion Based on CNNs and Transformers

CNNs have been widely applied in HSI-LiDAR joint classification owing to their powerful spatial feature extraction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

capability. Hang et al. [36] proposed a coupled CNN architecture, and Xu et al. [37] designed an end-to-end CNN framework. However, CNNs are constrained by local receptive fields, making it difficult to capture long-range dependencies. Paoletti et al. [8] highlighted the limitations of CNNs in complex scenarios in their review.

To overcome this deficiency, the transformer architecture has been introduced into the remote sensing domain [10], [11]. Ding et al. [12] proposed GLT-Net to capture multiscale dependencies by combining local and global attention. Zhao et al. [13] developed HCT, which integrates the advantages of CNNs and transformers. Yao et al. [14] introduced ExViT, providing a novel framework for multimodal classification. Feng et al. [15] designed DSHFNet, which achieves dynamic scale fusion based on multiattention mechanisms. Roy et al. [16] developed cross-HL to model cross-modal correlations through cross-attention. Feng et al. [17] proposed S^2EFT, achieving comprehensive fusion of spectral–spatial-elevation features. Zhu et al. [18] proposed a hierarchical multimodal feature aggregation method based on multihead axial attention transformers, effectively reducing computational complexity. Despite the demonstrated advantages of transformers, their $O(N^2)$ complexity limits their application on large-scale data. SSMs and Mamba.

B. Graph Neural Networks and Attention Mechanisms

Graph neural networks, as an alternative modeling paradigm, have been explored in remote sensing classification. Hong et al. [38] proposed graph convolutional networks for hyperspectral classification. Cai et al. [21] designed GAMF, which explores the correlation and complementarity between HSI and LiDAR. Attention mechanisms play a crucial role in feature fusion. CBAM proposed by Woo et al. [34] and the SE module designed by Hu et al. [35] demonstrated the effectiveness of channel and spatial attention. Although these methods have achieved progress in feature representation, there remains room for improvement in multiscale collaborative modeling and computational efficiency.

C. Mamba Architecture and SSMs

SSMs, particularly the Mamba architecture, as shown in Fig. 1, provide a new paradigm for efficient sequence modeling [22], [23]. Mamba, proposed by Gu and Dao [22], achieves effective modeling while maintaining $O(N)$ linear complexity through selective state space mechanisms. In the remote sensing domain, Chen et al. [24] proposed ChangeMamba for change detection, and Ma et al. [25] developed RS3Mamba for semantic segmentation.

MSFMamba proposed by Gao et al. [26] achieves multiscale feature fusion through hierarchical state space modeling, validating the potential of Mamba in multiscale modeling. Liao et al. [27] first introduced Mamba into HSI-LiDAR joint classification, proposing the HLMamba method. Shi et al. [28] further developed TBi-Mamba, employing bidirectional Mamba blocks to capture bidirectional dependencies. However, these methods primarily adopt simple parallel processing strategies, exhibiting deficiencies in adaptive weight allocation and multiscale collaborative modeling.

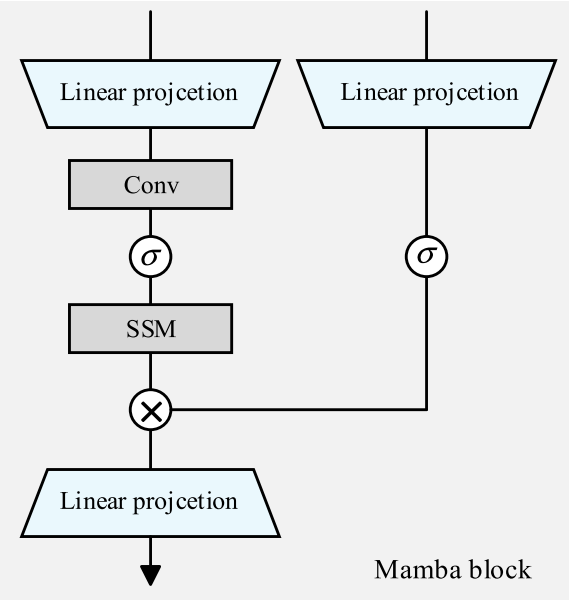


Fig. 1. Structure of Mamba.

D. Self-Supervised Learning and Multiscale Fusion

Beyond supervised learning, self-supervised learning provides new pathways for scenarios with limited labeled data. S^2MAE proposed by Lin et al. [29] learns robust features through masked prediction, demonstrating excellent performance in few-shot scenarios. NNCL introduced by Wang et al. [30] enhances feature discriminability by leveraging nearest neighbor samples. PICN developed by Gao et al. [31] compensates for information loss through prototype learning.

Multiscale feature fusion is an important strategy for improving classification performance. FPN proposed by Lin et al. [32] achieves multiscale fusion through a top-down pathway, and DenseNet designed by Huang et al. [33] enhances feature propagation through dense connections. However, existing multiscale methods predominantly employ simple fusion strategies, lacking effective cross-scale interaction mechanisms and exhibiting limitations when processing heterogeneous multimodal data.

III. Methodology

This section presents the HMMamba network architecture for HSI-LiDAR joint classification. HSI-LiDAR joint classification faces three main challenges: existing fusion methods employ fixed weighting strategies that fail to adapt to the heterogeneous importance of different modalities across land cover types; although multiscale features are utilized, most methods lack effective cross-scale interaction mechanisms, limiting their ability to capture optimal discriminative information at different observation levels; while global context modeling is essential, CNNs suffer from limited receptive fields and transformers face quadratic computational complexity, making it difficult to balance efficiency and capability.

To address these challenges, HMMamba achieves efficient multimodal multiscale feature fusion through four core

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

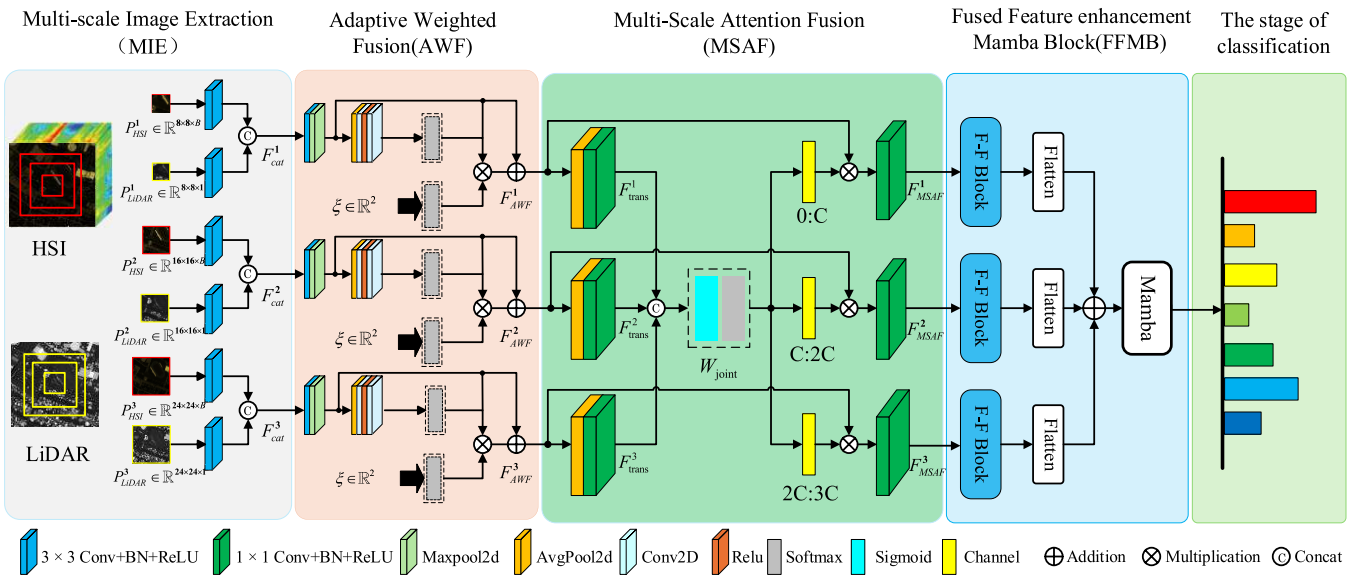


Fig. 2. Overall architecture diagram of the proposed method. This method contains five core modules: MIE, AWF, MSAF, FFMB, and a classification module, achieving effective fusion and classification of HSI and LiDAR data.

modules (see Fig. 2): multiscale image extraction (MIE) captures features at three spatial scales; AWF dynamically balances modal importance through hierarchical weight control; MSAF selects optimal observation scales via competitive attention mechanisms; and FFMB combines dual-branch attention with Mamba's linear-complexity sequence modeling for efficient global dependency capture.

Let $X_{\text{HSI}} \in \mathbb{R}^{H\times W\times B}$ and $X_{\text{LiDAR}} \in \mathbb{R}^{H\times W\times 1}$ denote the HSI and LiDAR data, where H, W, and B represent the image height, width, and number of spectral bands, respectively. For each target pixel to be classified, we extract its surrounding spatial neighborhoods at three scales with patch sizes $(P_1, P_2, P_3) = (8, 16, 24)$ pixels, yielding multiscale HSI-LiDAR patch pairs $\{P^s_{\text{HSI}} \in \mathbb{R}^{P_s\times P_s\times B}, P^s_{\text{LiDAR}} \in \mathbb{R}^{P_s\times P_s\times 1}$ for $s \in \{1, 2, 3\}$.

Different land cover types exhibit distinct feature representations across spatial scales. Fine-scale patches (8×8) primarily capture local texture information, suitable for identifying small objects (e.g., individual trees and vehicles) occupying 5–10 pixels, where high-frequency detail features (e.g., edges and corners) are critical for discrimination. Medium-scale patches (16×16) capture mid-level semantic features, appropriate for medium-sized objects (e.g., buildings and small agricultural fields) spanning 15–30 pixels, balancing local details with spatial structural information. Coarse-scale patches (24×24) provide global contextual information, essential for large-scale spatial patterns (e.g., building clusters and road networks) by capturing interobject spatial relationships and overall layout through larger receptive fields. Single-scale patches cannot simultaneously provide optimal discriminative features for all categories, necessitating our hierarchical multiscale strategy.

A. Multiscale Feature Extraction Module

For each scale s, HSI and LiDAR data are processed through independent convolutional encoders to extract modality-specific features

$$F^s_{\text{HSI}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times 3}\left(P^s_{\text{HSI}}\right)\right)\right) \in \mathbb{R}^{N_b\times C\times P_s\times P_s} \quad (1)$$

where N_b denotes batch size, $C = 64$ is the output channel dimension, $\text{Conv}_{3\times 3}$ represents a 3×3 convolutional layer (stride = 1 and padding = 1), BN denotes batch normalization, and ReLU is the activation function

$$F^s_{\text{LiDAR}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times 3}\left(P^s_{\text{LiDAR}}\right)\right)\right) \in \mathbb{R}^{N_b\times C\times P_s\times P_s}. \quad (2)$$

With the same parameter settings as (1).

The separate encoding design preserves modality-specific characteristics before fusion, allowing each encoder to learn optimal representations for spectral (HSI) and geometric (LiDAR) information without premature interference. The two modalities are then concatenated along the channel dimension

$$F^s_{\text{cat}} = \left[F^s_{\text{HSI}}; F^s_{\text{LiDAR}}\right] \in \mathbb{R}^{N_b\times 2C\times P_s\times P_s} \quad (3)$$

where [;] denotes channel-wise concatenation, resulting in 128 channels (64 from HSI and 64 from LiDAR).

This initial concatenation creates a joint representation that will be further optimized by the subsequent AWF module for adaptive cross-modal fusion.

B. AWF Module

Traditional multimodal fusion methods employ fixed weighting strategies (e.g., equal weights $\alpha_{\text{HSI}} = \alpha_{\text{LiDAR}} = 0.5$ or manually tuned ratios) that fail to adapt to the heterogeneous importance of HSI and LiDAR across different land cover types. Empirical observations reveal scene-dependent modality preferences: vegetation classification heavily relies on HSI's rich spectral signatures (chlorophyll absorption at 680 nm, near-infrared plateau at 750–1300 nm), while building discrimination depends primarily on LiDAR's elevation information for distinguishing multistory structures from ground-level objects. This scene-dependent modality importance cannot

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

be captured by static fusion, leading to suboptimal feature representations.

To address this limitation, we propose an adaptive weight allocation mechanism that decomposes the fusion problem into two complementary components: global learnable parameters ξ encode dataset-level modal priors learned across the entire training set (e.g., "vegetation-rich datasets generally favor HSI"); while an input-dependent gating network performs sample-level dynamic adjustment based on local feature statistics (e.g., "this specific pixel exhibits strong elevation variation → increase LiDAR weight"). This hierarchical design enables the model to learn both population-level tendencies and instance-specific adaptations, significantly improving upon fixed-weight approaches while maintaining computational efficiency through parameter sharing.

For each scale s, the concatenated features first undergo deep feature fusion and downsampling through a convolutional module

$$F^s_{\text{fused}} = \text{MaxPool}_{2\times2}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}\left(F^s_{\text{cat}}\right)\right)\right)\right) \in \mathbb{R}^{N_b\times64\times H_s\times W_s} \quad (4)$$

where $H_s = W_s = P_s/2$. The 3×3 convolution reduces channel dimension from 128 to 64 while integrating cross-modal information, and the 2×2 max pooling (stride = 2) downsamples spatial dimensions by half. The feature dimensions after processing are: $F^1_{\text{fused}} \in \mathbb{R}^{N_b\times64\times4\times4}$, $F^2_{\text{fused}} \in \mathbb{R}^{N_b\times64\times8\times8}$, and $F^2_{\text{fused}} \in \mathbb{R}^{N_b\times64\times12\times12}$.

We then apply global average pooling to capture the global context, followed by two consecutive 1×1 convolutional layers with channel reduction to generate the gating weights

$$G^s = \text{GAP}\left(\mathbf{F}^s_{\text{fused}}\right) \in \mathbb{R}^{N_b\times64\times1\times1} \quad (5)$$

where GAP denotes global average pooling that aggregates spatial information by averaging over all spatial positions

$$A^s = \text{ReLU}\left(\text{Conv}^{(1)}_{1\times1}(G^s)\right) \in \mathbb{R}^{N_b\times32\times1\times1} \quad (6)$$

where $\text{Conv}^{(1)}_{1\times1}$ reduces the channel dimension from 64 to 32

$$\left[\alpha^s_H, \alpha^s_L\right] = \text{Softmax}\left(\text{Conv}^{(2)}_{1\times1}(A^s)\right) \in \mathbb{R}^{N_b\times2\times1\times1} \quad (7)$$

where α^s_H and α^s_L denote the local gating weights for HSI and LiDAR features, respectively. $\text{Conv}^{(2)}_{1\times1}$ expands dimension from 32 to 2 (one for each modality), and Softmax normalization ensures $\alpha^s_H + \alpha^s_L = 1$, enforcing competitive selection where increasing one modality's weight automatically decreases the other's.

Simultaneously, we introduce global learnable parameters to encode the average importance of HSI and LiDAR across the entire dataset

$$\xi_g = [\xi_H, \xi_L] = \text{Softmax}([\xi_H, \xi_L]) \in \mathbb{R}^2 \quad (8)$$

where $\xi_H, \xi_L > 0$ are initialized to 1.0 and automatically learned through backpropagation during training.

The AWF module combines the global prior ξ_g with the local posterior $[\alpha^s_H, \alpha^s_L]$

$$\mathbf{F}^s_{\text{AWF}} = \left(1 + \xi_H \cdot \alpha^s_H\right) \odot \mathbf{F}^s_{\text{HSI}} + \left(1 + \xi_L \cdot \alpha^s_L\right) \odot \mathbf{F}^s_{\text{LiDAR}} \in \mathbb{R}^{N_b\times64\times P_s\times P_s} \quad (9)$$

where $\odot$ denotes element-wise multiplication, and the scalar gating weights are spatially broadcast to match the dimensions of F^s_{HSI} and F^s_{LiDAR}.

The constant term 1 in (9) serves a critical role beyond mathematical convenience. During early training when learned weights ($\xi \cdot \alpha$) may be unstable or near-zero, the identity path ($\times1$) ensures stable gradient flow, preventing catastrophic feature collapse. This design draws inspiration from residual connections but applies the concept to modal weighting rather than layer stacking. Without this term, zero-initialized weights would block all gradient propagation, causing training failure. In addition, the multiplicative formulation ($1 + \xi \cdot \alpha$) rather than additive ($F + \xi \cdot \alpha \cdot F$) provides stronger modulation: when $\xi \cdot \alpha \approx 1$, features are doubled, creating more pronounced modal selection effects compared to additive approaches.

C. MSAF Module

After AWF processing, the multiscale features $\{F^{\mathbf{1}}_{\mathbf{AWF}}, F^{\mathbf{2}}_{\mathbf{AWF}}, F^{\mathbf{3}}_{\mathbf{AWF}}\}$ have achieved effective cross-modal fusion. However, different land cover classes exhibit optimal discriminability at different observation scales: narrow roads are best identified at fine scales (4×4) where lane markings are visible, while forest patches require coarse scales (12×12) to capture homogeneous canopy structure. MSAF implements adaptive scale selection through a competitive attention mechanism that automatically learns the most discriminative scale(s) for each land cover type. Since the three scales have different spatial resolutions (4×4, 8×8, and 12×12), global average pooling first normalizes spatial dimensions

$$F^s_{\text{pool}} = \text{GAP}\left(F^s_{\text{AWF}}\right) \in \mathbb{R}^{N_b\times64\times1\times1}, \quad s \in \{1,2,3\}. \quad (10)$$

After pooling, all scales share identical dimensions, enabling fair comparison. To enhance scale-specific discriminability, independent nonlinear transformations are applied

$$F^s_{\text{trans}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(F^s_{\text{pool}}\right)\right)\right) \in \mathbb{R}^{N_b\times64\times1\times1} \quad (11)$$

where each scale employs separate convolutional parameters, allowing the network to learn scale-specific patterns (high-frequency for fine scales and low-frequency for coarse scales).

The features from the three scales are concatenated and processed through dual activation

$$F_{\text{concat}} = \left[F^1_{\text{trans}}; F^2_{\text{trans}}; F^3_{\text{trans}}\right] \in \mathbb{R}^{N_b\times192\times1\times1} \quad (12)$$

$$W_{\text{joint}} = \text{Softmax}\left(\text{Sigmoid}\left(\text{Conv}_{1\times1}\left(F_{\text{concat}}\right)\right)\right) \in \mathbb{R}^{N_b\times192\times1\times1} \quad (13)$$

where Sigmoid compresses magnitudes to [0, 1] and Softmax enforces competitive selection with $\sum_{c=1}^{192} W_c = 1$.

The joint weights are decomposed into scale-specific attention

$$W^s = W_{\text{joint}}[:, (s-1)\times64 : s\times64, :, :] \in \mathbb{R}^{N_b\times64\times1\times1}. \quad (14)$$

Channel-wise attention is applied with spatial broadcasting

$$F^s_{\text{att}} = W^s \odot F^s_{\text{AWF}} \in \mathbb{R}^{N_b\times64\times H_s\times W_s} \quad (15)$$

where W^s is the broadcast from $1\times1\times64$ to $H_s\times W_s\times64$ (with $H_s = W_s \in \{4, 8, 12\}$). A projection layer refines the features

$$F^s_{\text{MSAF}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}^{64\to64}_{1\times1}\left(F^s_{\text{att}}\right)\right)\right) \in \mathbb{R}^{N_b\times64\times H_s\times W_s} \quad (16)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

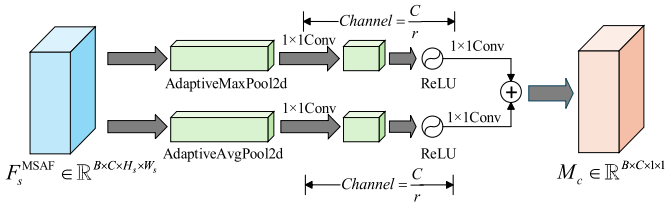


Fig. 3. Structure of FFMB for channel branch.

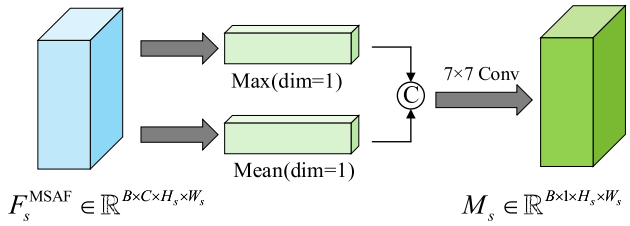


Fig. 4. Structure of FFMB for spatial branch.

where the 1×1 convolution maintains 64 channels, followed by batch normalization and ReLU activation to enhance feature nonlinearity. The competitive attention mechanism in MSAF automatically suppresses less informative scales, preventing redundant multiscale processing while enabling the model to discover optimal scale preferences for different land cover types.

D. FFMB Module

After AWF and MSAF process multiscale features through cross-modal and cross-scale fusion, efficiently modeling global spatial dependencies remains a critical challenge. This challenge stems from three fundamental limitations in existing approaches: 1) limited receptive fields: CNNs rely on small local kernels (3×3 or 5×5), requiring many stacked layers to capture large-scale structures such as road networks or agricultural fields, which leads to training difficulties and excessive parameters; 2) quadratic complexity: transformer self-attention achieves global coverage through pairwise similarity computation, but incurs $O(N^2)$ complexity that becomes computationally prohibitive for high-resolution patches; and 3) inadequate channel-spatial modeling: neither CNNs NOR transformers explicitly capture the coupled relationships between spectral channels (encoding material properties like chlorophyll absorption at 680 nm) and spatial patterns (encoding geometric structures like building rectangles), which are essential for HSI-LiDAR classification where different land cover types require varying emphasis on spectral versus spatial information.

FFMB addresses these limitations through a unified pipeline combining dual-branch attention, multiscale integration, and selective state space modeling. The module processes features at each scale s independently, taking as input the fused multimodal feature $F_s^{\text{MSAF}} \in \mathbb{R}^{N_b \times C \times H_s \times W_s}$ from MSAF, where N_b denotes batch size, $C = 64$ is the channel dimension, and (H_s, W_s) are scale-dependent spatial dimensions.

1) Dual-Branch Attention: The attention mechanism separates feature refinement into two complementary tasks: channel selection (identifying informative spectral-elevation bands) and spatial localization (highlighting discriminative regions). This design exploits the distinct nature of HSI-LiDAR data—spectral channels encode material signatures (e.g., NIR reflectance for vegetation, elevation for building height), while spatial patterns encode geometric configurations (e.g., rectangular footprints and linear networks).

For the channel branch is shown in Fig. 3, we aggregate spatial information using two complementary pooling operations

$$F_{\max}^{c} = \text{AdaptiveMaxPool2d}\left(F_s^{\text{MSAF}}\right) \in \mathbb{R}^{N_b \times C \times 1 \times 1} \tag{17}$$

$$F_{\text{avg}}^{c} = \text{AdaptiveAvgPool2d}(F_s^{\text{MSAF}}) \in \mathbb{R}^{N_b \times C \times 1 \times 1}. \tag{18}$$

Max pooling extracts peak responses critical for material identification (e.g., maximum NIR reflectance at 850 nm for healthy vegetation and peak elevation for tall structures), while average pooling captures central tendencies characteristic of homogeneous regions (e.g., consistently low NIR for water and stable elevation for flat terrain). Both descriptors are then processed through a shared squeeze-and-excitation pathway with a bottleneck design

$$M_c^{\max} = \text{Conv}_{1\times1}^{C}\left(\text{ReLU}\left(\text{Conv}_{1\times1}^{C/r}\left(F_{\max}^{c}\right)\right)\right) \tag{19}$$

$$M_c^{\text{avg}} = \text{Conv}_{1\times1}^{C}\left(\text{ReLU}\left(\text{Conv}_{1\times1}^{C/r}\left(F_{\text{avg}}^{c}\right)\right)\right) \tag{20}$$

$$M_c = M_c^{\max} + M_c^{\arg} \in \mathbb{R}^{N_b \times C \times 1 \times 1} \tag{21}$$

where $r = 16$ is the compression ratio. The bottleneck architecture: first reducing channels to C/r, then expanding back to C— enforces compact channel correlation learning with $2C^2/r$ parameters (versus $4C^2$ for independent branches), while summation enables both statistics to contribute complementary information.

Complementary to channel attention, the spatial branch identifies discriminative regions by pooling information across channels is shown in Fig. 4

$$F_{\max}^{s} = \max_{c=1}^{c}\left(F_s^{\text{MSAF}}\right) \in \mathbb{R}^{N_b \times 1 \times H_s \times W_s} \tag{22}$$

$$F_{\text{avg}}^{s} = \frac{1}{C}\sum_{c=1}^{C}\left(F_s^{\text{MSAF}}\right) \in \mathbb{R}^{N_b \times 1 \times H_s \times W_s} \tag{23}$$

$$M_s = \text{Conv}_{7\times7}\left(\left[F_{\max}^{s}; F_{\text{avg}}^{s}\right]\right) \in \mathbb{R}^{N_b \times 1 \times H_s \times W_s}. \tag{24}$$

Cross-channel max pooling emphasizes locations with strong responses in any band (e.g., class boundaries with sharp spectral-elevation transitions and distinct objects like vehicles), while average pooling represents overall multispectral characteristics (e.g., homogeneous forest canopies and background context). The 7×7 convolution captures diagnostic spatial patterns spanning 5–10 pixels (edges, corners, and textures) while maintaining efficiency—at 10-m resolution, the kernel covers approximately 50×50 m, sufficient for most land cover objects without requiring excessive parameters.

Adaptive fusion is shown in Fig. 5. Rather than using fixed weights, we employ a learnable fusion strategy that dynamically balances channel and spatial attention based on scene content. This is essential because different land covers exhibit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

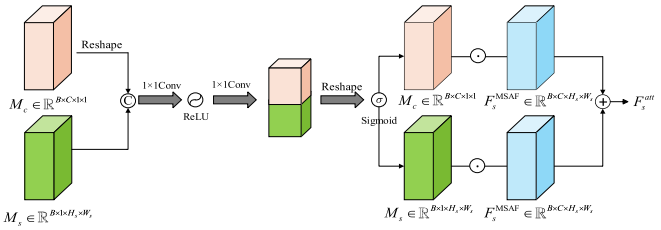


Fig. 5. Structure of FFMB for adaptive fusion.

different spectral–spatial dependencies—forests rely heavily on spectral discrimination (NIR/red ratio for NDVI), while urban structures depend on geometric patterns (rectangular shapes and elevation boundaries).

The channel attention $M_c \in \mathbb{R}^{N_b \times C \times 1 \times 1}$ is maintained compact, while spatial attention $M_s \in \mathbb{R}^{N_b \times 1 \times H_s \times W_s}$ is reshaped to $M_s^{\text{flat}} \in \mathbb{R}^{N_b \times H_s W_s \times 1 \times 1}$, treating each location as an independent "channel." They are then concatenated

$$F_{\text{concat}} = \left[M_c; M_s^{\text{flat}}\right] \in \mathbb{R}^{N_b \times (C + H_s W_s) \times 1 \times 1}. \tag{25}$$

An adaptive MLP processes this joint representation to generate branch-specific weights

$$W_{\text{fusion}} = \text{MLP}(F_{\text{concat}}) \in \mathbb{R}^{N_b \times (C + H_s W_s) \times 1 \times 1} \tag{26}$$

where MLP follows a bottleneck design: $\text{Conv}_{1\times1}^{(C+H_sW_s)/r} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{1\times1}^{C+H_sW_s}$ with $r = 16$. Since the joint dimension $(C + H_sW_s)$ varies across scales, the MLP is dynamically reinstantiated during each forward pass based on the input spatial resolution, allowing the same dual-branch attention module to process different scales sequentially. The weights are then decomposed and activated

$$W_c = \sigma(\text{Reshape}(W_{\text{fusion}}[:, :C, :, :])) \in \mathbb{R}^{N_b \times C \times 1 \times 1} \tag{27}$$

$$W_s = \sigma\left(\text{Reshape}\left(W_{\text{fusion}}[:, C:, :, :]\right)\right) \in \mathbb{R}^{N_b \times 1 \times H_s \times W_s} \tag{28}$$

where the channel weight reshape maintains the channel dimension $(C, 1, 1)$, and the spatial weight reshape converts the flattened spatial dimension back to $(1, H_s, W_s)$. The Sigmoid activation σ constrains weights to [0, 1]. The asymmetric dimensions (C channel weights versus $H_s \times W_s$ spatial weights) enable independent control over spectral band selection and spatial region emphasis. The enhanced feature is computed via additive fusion

$$F_s^{\text{att}} = F_s^{\text{MSAF}} \odot W_c + F_s^{\text{MSAF}} \odot W_s. \tag{29}$$

This formulation allows complementary enhancement: W_c modulates spectral discriminability (emphasizing informative bands like NIR for vegetation), while W_s highlights geometric saliency (focusing on rooftops or road centerlines), with both contributions independently weighted. The additive design provides robustness during training—when one branch produces weak attention, the other can still contribute, unlike multiplicative fusion which requires simultaneous activation of both branches.

2) Multiscale Integration: The three attention-enhanced features $\{F_s^{\text{att}}\}_{s=1}^3$ with different spatial sizes (4×4, 8×8, and 12×12) are first spatially flattened, then transformed via scale-specific linear layers to a unified sequence length N

$$\tilde{F} = \text{Flatten}\left(F_s^{\text{att}}\right) \in \mathbb{R}^{N_b \times C \times H_s W_s} \tag{30}$$

$$E_s = \text{Linear}_s\left(\tilde{F}_s\right) \in \mathbb{R}^{N_b \times C \times N}, \quad s = 1, 2, 3 \tag{31}$$

where $N = P^2$ is the target sequence length (e.g., $N = 64$ for $P = 8$). The learnable transformations enable task-optimized unification—fine scales preserve high-frequency details while coarse scales retain low-frequency patterns. The unified features are summed and transposed for sequence modeling

$$F^{\text{fused}} = (E_1 + E_2 + E_3)^{\text{T}} \in \mathbb{R}^{N_b \times N \times C} \tag{32}$$

where the transpose converts channel-major to sequence-major format for subsequent Mamba processing.

3) Mamba Sequence Modeling: After multiscale integration, the fused feature $F^{\text{fused}} \in \mathbb{R}^{N_b \times N \times C}$ encodes rich spectral–spatial-elevation information across different observation scales. However, effective HSI-LiDAR classification requires modeling long-range spatial dependencies that are critical for discriminating large-scale land cover patterns—road networks spanning hundreds of pixels, agricultural field boundaries extending across multiple patches, and forest-urban transitions requiring global contextual understanding. Traditional CNNs with local receptive fields cannot capture these extended spatial relationships without excessive depth, while transformer self-attention incurs $O(N^2)$ computational complexity that becomes prohibitive for high-resolution remote sensing data. To address this challenge, we employ a selective SSM (Mamba) that achieves global dependency modeling at $O(N)$ linear complexity through content-aware sequential processing

$$\hat{F}^{\text{fused}} = \text{LayerNorm}\left(F^{\text{fused}}\right) \tag{33}$$

$$X_{\text{ssm}} = \text{Linear}_{\text{ssm}}\left(\hat{F}^{\text{fused}}\right) \in \mathbb{R}^{N_b \times N \times D} \tag{34}$$

$$X_{\text{gate}} = \text{Linear}_{\text{gate}}\left(\hat{F}^{\text{fused}}\right) \in \mathbb{R}^{N_b \times N \times D} \tag{35}$$

where $D = 2C = 128$ is the expanded dimension. Following Mamba's gated architecture, X_{ssm} enters the selective state space for sequential processing, while X_{gate} provides multiplicative gating control. The key innovation lies in content-dependent parameter generation: for each sequence position t (corresponding to a spatial location in the original feature map), three input-specific parameters dynamically adapt to local land cover characteristics

$$\Delta_t = \text{softplus}\left(\text{Linear}_\Delta(x_t)\right) \tag{36}$$

$$B_t = \text{Linear}_B(x_t), \quad C_t = \text{Linear}_C(x_t) \tag{37}$$

where $x_t = X_{\text{ssm}}[:, t, :]$. These parameters enable adaptive processing tailored to different land cover types: Δ_t (time step) controls the temporal discretization—smaller values preserve long-range context for modeling extended structures like road networks or irrigation systems, while larger values enable rapid local adaptation for isolated objects like individual buildings or vehicles; B_t (input gate) determines how strongly current spectral-elevation observations influence state

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

updates—discriminative pixels at class boundaries (vegetation-soil transitions and building edges) receive larger B_t for strong feature incorporation, while homogeneous regions (uniform water bodies, continuous forest canopies) use smaller B_t to avoid redundant information accumulation; C_t (output gate) selectively emphasizes state dimensions encoding task-relevant patterns, suppressing background noise while highlighting material-specific spectral signatures and geometric structures.

The hidden state evolves sequentially to accumulate spatial context

$$h_t = \exp(\Delta_t \cdot A) \odot h_{t-1} + (\Delta_t \cdot B_t) \odot x_t \tag{38}$$

$$y_t = C_t \odot h_t \tag{39}$$

where A is a learnable transition matrix controlling memory decay across spatial positions. This recurrent formulation enables progressive context aggregation—early positions in the sequence accumulate local spectral-elevation patterns, while later positions integrate this information with global land use context to form comprehensive scene understanding. For instance, classifying a pixel as "urban road" requires both local cues (asphalt spectral signature and flat elevation) and global context (surrounded by buildings and part of a linear network), which the state evolution naturally captures through sequential integration.

The SSM output is combined with gating and projected back

$$Z = X_{\text{gate}} \odot \text{SiLU}(Y_{\text{ssm}}) \tag{40}$$

$$F^{\text{out}} = \text{Linear}_{\text{proj}}(Z) + F^{\text{fused}} \in \mathbb{R}^{N_b \times N \times C} \tag{41}$$

where $Y_{\text{ssm}} = [y_1, \ldots, y_N]$ and the residual connection preserves gradient flow. The critical advantage for HSI-LiDAR classification is computational efficiency: processing N positions requires $O(N)$ operations versus $O(N^2)$ for self-attention, enabling efficient modeling of global spatial relationships in high-resolution multimodal remote sensing imagery while maintaining linear complexity. The complete Mamba processing flow is shown in Algorithm 1.

E. Classification Stage

The final classification is achieved through a lightweight CNN classifier for computational efficiency and overfitting prevention. The sequence features output from the FFMB module undergo a dimension transformation

$$x_{\text{img}} = \text{Reshape}(y_{\text{mamba}}, [N_b, 64, P, P]) \in \mathbb{R}^{N_b \times 64 \times P \times P}. \tag{42}$$

The classifier consists of two 1×1 convolutional layers with adaptive global average pooling

$$F_{\text{cls}} = \text{AvgPool2d}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}^{64\times32}(x_{\text{img}})\right)\right)\right) \in \mathbb{R}^{N_b \times 32 \times 1 \times 1} \tag{43}$$

$$y = \text{Softmax}\left(\text{Conv}_{1\times1}^{32\to N_C}(F_{\text{cls}})\right) \in \mathbb{R}^{N_b \times N_c} \tag{44}$$

where N_c denotes the number of classes. AdaptiveAvgPool2d ensures spatial dimension invariance: regardless of input size, the output is always 1×1, improving classification robustness to spatial variations. Through the collaborative operation of the above modules, HMMamba effectively extracts and fuses multimodal multiscale features while utilizing Mamba's linear complexity advantages to achieve efficient global dependency modeling.

IV. Experimental Results and Analysis

A. Dataset Description

To comprehensively validate the effectiveness and generalization ability of the proposed method, experiments are conducted on four publicly available datasets with different geographical environmental characteristics and classification complexity: Houston2013, MUUFL, Trento, and Augsburg. These datasets cover diverse scenarios, including urban, rural, and campus environments, with different spatial resolutions, spectral characteristics, and land cover types.

For HSI data preprocessing, bands heavily affected by water vapor absorption (1.35–1.43 μm and 1.80–1.95 μm) were removed to reduce atmospheric interference and noise, retaining only effective bands with higher signal-to-noise ratios. LiDAR data were normalized using min-max scaling to the range [0, 1] to ensure numerical consistency with hyperspectral features. Spatial co-registration was performed using an affine transformation with subpixel accuracy (RMSE <0.5 pixels) to ensure precise alignment between modalities.

A stratified random sampling strategy was adopted for dataset partitioning, where a fixed number of samples from each class were randomly selected as the training set, and the remaining samples were used as the test set. To ensure experimental reproducibility, the same random seed was used for data partitioning across all experiments.

1) Houston2013 Dataset: Collected from the University of Houston campus and surrounding urban areas, this dataset consists of HSI with 349 × 1905 pixels at 2.5-m spatial resolution, covering 0.38–1.05 μm with 144 bands. It provides 15 029 labeled samples across 15 urban land cover classes, with 20 samples per class for training (300 total) and 14 729 for testing.

2) MUUFL Dataset: Acquired from the University of Southern Mississippi Gulfport campus, containing 325 × 220 pixels covering 0.38–1.05 μm with 64 effective bands (preprocessed from 72). The dataset contains 53 687 labeled samples divided into 11 land cover classes, with 220 training and 53 467 testing samples.

3) Trento Dataset: Collected from rural areas in southern Trento, Italy, with 166 × 600 pixels at 1-m spatial resolution, wavelength range 0.42–0.99 μm, and 63 spectral bands. It contains 30 214 labeled samples covering 6 agricultural classes, with 120 training and 30 094 testing samples.

4) Augsburg Dataset: From Augsburg, Germany, resampled to 30-m spatial resolution with dimensions 332 × 485 pixels. This dataset provides 78 294 labeled samples across 7 urban land cover classes, with 140 training and 78 154 testing samples. The detailed class information of the four datasets is shown in Tables I and II.

B. Experimental Setup

All experiments are implemented using PyTorch 2.1.0 on a server equipped with an Intel Core i9-14900KF CPU and NVIDIA GeForce RTX 4090D GPU (24-GB memory).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

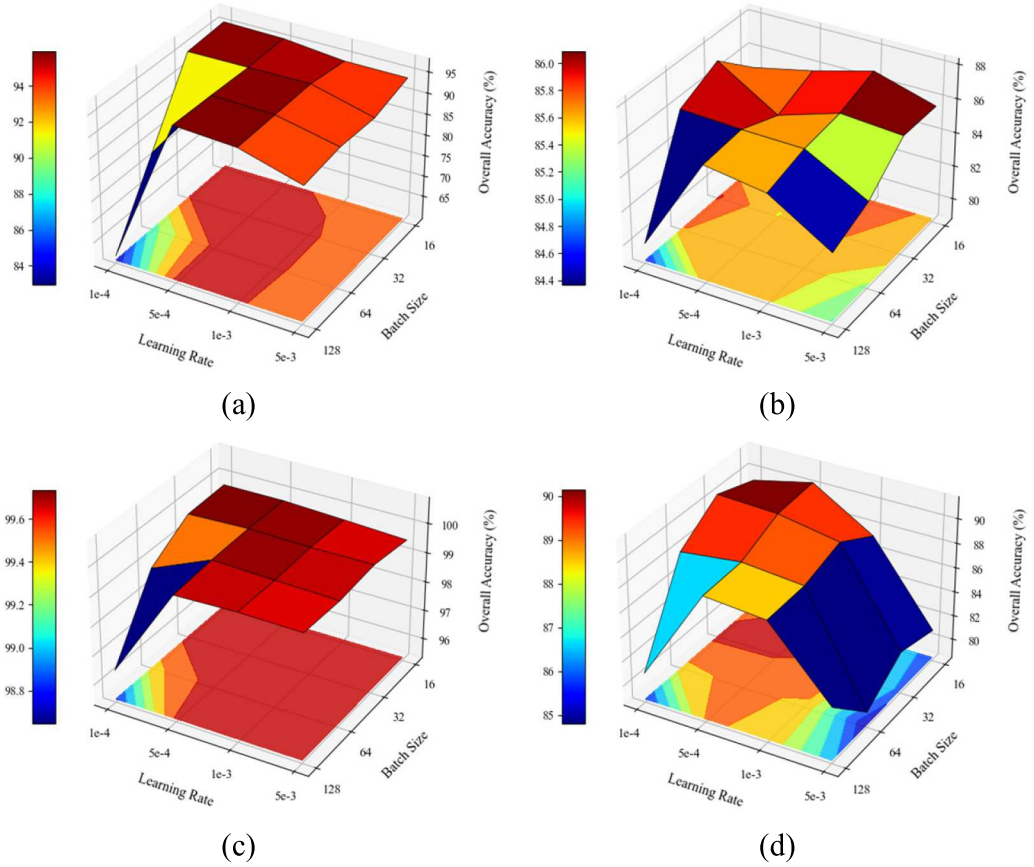


Fig. 6. Impact of learning rate and batch size on performance. (a) Houston2013 dataset. (b) MUUFL dataset. (c) Trento dataset. (d) Augsburg dataset.

The Adam optimizer is employed with $\beta = 0.9$, weight decay = 5e−4. Maximum training epochs are set to 500 with early stopping (patience = 50 epochs) when validation accuracy plateaus. Multiscale inputs use spatial sizes of 8×8, 16×16, and 24×24 pixels.

For fair comparison, all baseline methods use parameters from their original papers. Methods requiring different input sizes are adapted using appropriate padding or cropping. When official implementations are available (GLT, HCT, DSHF, and HLMamba), we use the authors' code; otherwise, we carefully reimplemented methods following published descriptions. All results are averaged over 10 runs with different random seeds to ensure statistical reliability.

Three evaluation metrics are employed: overall accuracy (OA), average accuracy (AA), and Kappa coefficient ($K \times 100$).

C. Parameter Analysis

Learning rate and batch size significantly affect model convergence and performance. We conduct grid search experiments with learning rates {1e−4, 5e−4, 1e−3, 5e−3} and batch sizes {16, 32, 64, 128} across all datasets. Fig. 6 presents the results as heatmaps, where red regions indicate the highest OA values and dark blue regions indicate the lowest values.

The optimal hyperparameters vary across datasets due to their intrinsic characteristics. For Houston2013, optimal performance occurs at a learning rate (5e−4 to 1e−3) with batch size 64. This dataset contains 15 complex urban classes with diverse spectral–spatial patterns, where moderate learning rates balance convergence speed and accuracy without overshooting optima.

MUUFL favors smaller learning rates (1e−4 to 5e−4) primarily due to severe class imbalance—yellow curb has only 183 total samples while trees has 23 246. Conservative learning rates prevent the model from being dominated by majority classes and ensure minority classes are adequately learned through more gradual weight updates.

Trento achieves best results with both a small learning rate (1e−4) and a small batch size (32). With only 6 agricultural classes exhibiting distinct spectral characteristics, the model can quickly identify obvious class separations. Small learning rates prevent overshooting these well-defined boundaries, while small batch sizes provide more frequent gradient updates to refine subtle within-class variations in homogeneous agricultural parcels.

Augsburg's low spatial resolution (30 m) necessitates conservative learning rates (1e−4 to 5e−4) to fully exploit limited spatial details. Coarse resolution provides less discriminative features, requiring the model to learn subtle spectral differences more carefully without aggressive parameter updates that might cause instability.

These dataset-specific characteristics explain the variation in optimal hyperparameters. To ensure fair comparison and practical applicability, we adopt a learning rate 5e−4 and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR HOUSTON2013 AND MUUFL DATASETS

NO	Houston2013			MUUFL		
	Class Name	Training	Test	Class Name	Training	Test
1	Healthy Grass	20	1231	Trees	20	23226
2	Stressed Grass	20	1234	Mostly Grass	20	4250
3	Synthetic Grass	20	677	Mixed Ground Surface	20	6862
4	Trees	20	1224	Dirt and Sand	20	1806
5	Soil	20	1222	Road	20	6667
6	Water	20	305	Water	20	446
7	Residential	20	1248	Buildings Shadow	20	2213
8	Commercial	20	1224	Buildings	20	6220
9	Road	20	1232	Sidewalk	20	1365
10	Highway	20	1207	Yellow Curb	20	163
11	Railway	20	1215	Cloth Panels	20	249
12	Parking Lot 1	20	1213			
13	Parking Lot 2	20	449			
14	Tennis Court	20	408			
15	Running Track	20	640			
-	Total	300	14729	Total	220	53467

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES IN THE TRENTO AND AUGSBURG DATASETS

NO	Trento			Augsburg		
	Class Name	Training	Test	Class Name	Training	Test
1	Apple Trees	20	4014	Forest	20	13487
2	Buildings	20	2883	Residential Area	20	30309
3	Ground	20	459	Industrial Area	20	3831
4	Woods	20	9103	Low Plants	20	26837
5	Vineyard	20	10481	Allotment	20	555
6	Roads	20	3154	Commercial Area	20	1625
7				Water	20	1510
-	Total	120	30094	Total	140	78154

TABLE III
IMPACT OF DIFFERENT MODULES ON RESULTS (BEST RESULTS DISPLAYED IN BOLD)

Case.	Component.			Dataset.					
	MIE+AWF	MSAF	FFMB	Houston2013	MUUFL	Trento	Augsburg	Params	FLOPs
1	√	×	×	94.81	84.72	99.48	87.75	111.86K	70.59M
2	√	√	×	95.58	85.06	99.54	88.01	137.21K	71.52M
3	√	×	√	95.64	85.01	99.57	88.29	145.11K	70.92M
4	√	√	√	**96.06**	**86.37**	**99.61**	**90.77**	**170.45K**	**71.85M**

batch size 64 as unified settings across all experiments. This configuration falls within the optimal range identified for each dataset, achieving consistent and competitive performance while maintaining computational efficiency and reproducibility.

D. Ablation Studies

To validate the effectiveness of each component, systematic ablation experiments were conducted on four datasets: Houston2013, MUUFL, Trento, and Augsburg. The independent contribution and synergistic effects of each component were evaluated by progressively adding different modules. The experimental results are presented in Table III.

The feature processing strategies for different configurations are as follows: Case 1 serves as the baseline, averaging the AWF outputs from three scales for direct classification; Case 2 introduces the MSAF module on top of the baseline, employing multiscale attention to weight the three-scale features for fusion, but without FFMB for global modeling, using the MSAF output directly for classification; Case 3 bypasses the MSAF scale attention processing, averaging the AWF outputs from three scales before feeding them into FFMB for global modeling; and Case 4 represents the complete

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

model incorporating all components, where AWF outputs are weighted by MSAF and subsequently modeled by FFMB.

1) Case 1 (Baseline—MIE+AWF): The baseline model employs MIE and AWF mechanisms. AWF achieves adaptive cross-modal fusion through dual-level weight control: global learnable parameters capture dataset-level modal importance trends, while the local gating network dynamically adjusts fusion weights according to each sample's characteristics. This hierarchical control strategy enables the model to automatically balance the contributions of HSI and LiDAR according to different land cover types, avoiding the limitations of fixed-weight fusion. The baseline model already demonstrates competitive performance, validating the effectiveness of the adaptive fusion strategy.

2) Case 2 (Adding MSAF): Upon introducing the MSAF module to the baseline, the OA on the four datasets improved by 0.77%, 0.34%, 0.06%, and 0.26%, respectively, while only adding 25.35K parameters and 0.93 M FLOPs (representing a 22.7% parameter increase and 1.3% computational increase compared to the baseline).

The performance improvement variations across different datasets reveal the scene-adaptive characteristics of MSAF.

Houston2013, as a complex urban scene containing multiscale objects ranging from small vehicles and roads to large buildings and green spaces, benefits from MSAF's competitive attention mechanism that selects the most discriminative scale features for different targets, achieving a notable 0.77% improvement. In contrast, the Trento and Augsburg datasets exhibit smaller improvements (0.06% and 0.26%, respectively), which can be attributed to their scene characteristics: Trento mainly consists of continuously distributed agricultural parcels with relatively uniform object scales, while Augsburg, despite being an urban scene, has its multiscale feature discriminability limited by the coarse 30-m resolution. The MUUFL dataset achieved a moderate improvement of 0.34%, reflecting the intermediate complexity of its campus environment.

3) Case 3 (Adding FFMB Without MSAF): This configuration bypasses MSAF and directly feeds the AWF outputs into FFMB for global modeling. Compared to the baseline, the OA on the four datasets improved by 0.83%, 0.29%, 0.09%, and 0.54%, respectively, with an increase of 33.25K parameters (29.7%) but only 0.33 M FLOPs (0.5%).

Notably, FFMB's improvements on the Houston2013 and Augsburg datasets (0.83% and 0.54%) exceed those of MSAF (0.77% and 0.26%), indicating the importance of global modeling in these complex scenes. Particularly on the Augsburg dataset, FFMB effectively compensates for the insufficient local discriminability caused by the 30-m coarse resolution by capturing broader spatial context through Mamba's global sequence modeling capability.

Furthermore, FFMB's dual-branch attention mechanism separately models channel correlations and spatial patterns, enabling simultaneous capture of spectral band relationships and spatial structural regularities. Its $O(N)$ linear computational complexity is significantly lower than MSAF's multiscale processing overhead, which explains why it achieves greater performance improvements on certain datasets while incurring lower computational costs.

4) Case 4 (Complete Model): The complete architecture achieves optimal performance across all datasets, with OA improvements of 1.25%, 1.65%, 0.13%, and 3.02% compared to the baseline. Importantly, the complete model exhibits different module interaction patterns across datasets.

On the MUUFL and Augsburg datasets, the complete model's gains (1.65% and 3.02%, respectively) significantly exceed the theoretical sum of Cases 2 and 3 (0.63% and 0.80%, respectively), demonstrating strong positive synergistic effects. For MUUFL, this can be attributed to the severe class imbalance in the dataset (trees class has 23 246 samples while yellow curb has only 163), where MSAF's scale selection helps FFMB better capture the spatial context of minority classes, and FFMB's global modeling guides MSAF to allocate more reasonable scale resources for different classes. For Augsburg, the global–local feature complementarity in the coarse-resolution environment produces strong enhancement effects between the two modules.

In contrast, on the Houston2013 and Trento datasets, the complete model's actual gains (1.25% and 0.13%, respectively) are slightly lower than the theoretical sums (1.60% and 0.15%, respectively), indicating some functional overlap between the two modules on these datasets. Nevertheless, the complete model still achieves optimal performance across all datasets, validating the effectiveness of end-to-end joint optimization among modules.

5) Parameter and Computational Efficiency Analysis: The complete model has a total of 170.45K parameters, with AWF accounting for 66% (111.86K), MSAF for 15% (25.35K), and FFMB for 19% (33.25K). Although the parameter count increased by 52.4% compared to the baseline, the total FLOPs only increased from 70.59 to 71.85 M (a 1.8% increase).

This parameter-computation efficiency advantage stems from three aspects of architectural design: 1) AWF's early fusion strategy unifies the feature channels of HSI and LiDAR to lower dimensions, reducing the number of features that subsequent modules need to process and avoiding dimensional explosion; 2) MSAF's competitive attention mechanism implements selective activation, activating only the most relevant scale branches for each input and avoiding redundant processing of all scale features; and 3) FFMB's Mamba architecture has $O(N)$ linear complexity, which significantly reduces computational costs when processing long sequences compared to the $O(N^2)$ quadratic complexity of traditional transformers, making global modeling feasible without introducing significant computational burden.

In summary, the ablation experiments validate the design rationale of the proposed method from multiple perspectives: AWF addresses cross-modal heterogeneity, MSAF handles multiscale feature selection, and FFMB achieves efficient global modeling. These three components work synergistically to significantly improve classification performance while maintaining computational efficiency.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON THE HOUSTON 2013 DATASET (BEST RESULTS DISPLAYED IN BOLD)

NO.	GLT (2022)	HCT (2023)	ExViT (2023)	DSHF (2023)	GAMF (2024)	S2ENet (2024)	CrossHL (2024)	HLMamba (2024)	GFSFN (2025)	Ours
C1	93.27	96.53	95.33	89.61	95.36	92.40	**97.80**	90.08	95.69	95.85
C2	95.71	98.49	97.92	98.13	98.46	94.51	97.48	98.03	**99.18**	97.32
C3	99.17	99.42	99.41	99.72	98.22	99.43	99.26	99.08	**99.70**	99.26
C4	98.77	94.28	96.64	**99.42**	94.28	96.18	93.70	97.00	94.28	96.41
C5	99.41	99.34	99.53	**100.00**	**100.00**	99.47	99.83	99.98	**100.00**	**100.00**
C6	95.99	99.52	96.25	99.61	98.03	97.11	95.73	95.34	**99.01**	**99.67**
C7	96.89	**99.03**	94.92	94.63	93.53	95.22	94.87	95.06	99.51	97.43
C8	86.30	79.90	83.39	82.92	69.61	85.62	77.04	87.49	81.69	**90.52**
C9	92.35	79.87	86.52	77.43	83.03	93.17	79.62	93.98	90.74	**96.13**
C10	92.14	97.25	91.26	92.62	**97.26**	91.61	94.11	94.31	97.76	96.11
C11	95.47	86.83	93.89	83.04	91.44	94.33	95.14	**97.64**	94.89	88.39
C12	88.96	92.91	87.20	98.35	94.64	87.51	96.04	90.97	91.50	**95.55**
C13	**98.61**	93.98	97.01	95.99	95.76	97.50	95.32	96.69	100.00	97.99
C14	**100.00**	**100.00**	99.26	**100.00**	**100.00**	99.80	100.00	**100.00**	**100.00**	**100.00**
C15	99.87	**100.00**	99.96	99.68	**100.00**	99.75	**99.68**	99.68	**100.00**	**100.00**
OA	94.78	93.55	93.60	92.87	92.89	94.01	93.54	95.12	95.42	**96.06**
AA	95.52	94.49	94.57	94.07	93.97	94.90	94.38	95.68	96.26	**96.71**
K	94.36	93.03	93.09	92.29	92.31	93.52	93.01	94.73	**95.05**	**95.73**

TABLE V
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON THE MUUFL DATASET (BEST RESULTS DISPLAYED IN BOLD)

NO.	GLT (2022)	HCT (2023)	ExViT (2023)	DSHF (2023)	GAMF (2024)	S2ENet (2024)	CrossHL (2024)	HLMamba (2024)	GFSFN (2025)	Ours
C1	87.87	84.41	87.41	79.98	92.41	90.15	88.98	93.09	88.15	**93.15**
C2	76.90	75.65	72.77	**82.95**	59.27	74.65	76.72	74.89	72.21	77.21
C3	60.67	60.75	64.13	68.13	**75.35**	64.09	49.63	52.96	65.38	68.49
C4	79.66	89.48	81.39	92.83	74.36	83.76	88.64	87.20	**97.51**	85.13
C5	81.59	73.59	79.54	**92.03**	88.18	80.73	77.62	72.05	80.38	78.86
C6	**100.00**	99.32	98.65	97.31	91.03	99.15	99.39	99.55	**100.00**	**100.00**
C7	88.46	85.90	71.84	89.65	86.48	84.47	78.98	88.23	88.45	**92.76**
C8	92.42	88.71	93.79	90.54	93.52	92.07	94.66	**96.38**	95.03	96.30
C9	66.61	50.54	63.58	22.52	52.74	58.95	56.70	53.77	55.97	**67.51**
C10	71.98	66.46	54.61	40.31	66.87	67.97	76.68	75.46	74.84	**88.34**
C11	96.38	97.99	95.98	98.39	99.59	95.42	98.46	95.58	99.59	**99.61**
OA	83.10	79.33	81.6	80.79	85.26	83.41	83.70	83.24	83.66	**86.37**
AA	82.04	79.35	78.52	77.69	79.98	81.04	81.31	80.83	83.41	**86.12**
K	77.58	73.61	76.27	75.67	80.75	78.54	78.68	78.18	79.01	**82.29**

E. Performance Comparison

We compared HMMamba against nine state-of-the-art methods: GLT [18], HCT [19], ExViT [45], DSHF [20], GAMF [43], S2ENet [44], CrossHL [21], HLMamba [11], and GFSFN [39]. These methods represent diverse architectures, including CNN-based, transformer-based, graph-based, and Mamba-based approaches. Tables IV–VII present the quantitative results across four datasets.

1) Houston2013 Results: HMMamba achieved 96.06% OA, 96.71% AA, and 95.73% Kappa coefficient, surpassing the second-best method GFSFN by 0.64%, 0.45%, and 0.68%, respectively. Our method obtained the highest accuracy in 7 out of 15 classes, achieving perfect classification (100%) for soil, tennis court, and running track. It attained 90.52% accuracy on the commercial class, significantly outperforming other methods, and reached 96.13% on the road class, demonstrating superior boundary delineation capability in complex urban scenarios. These improvements stem from FFMB's dual-branch attention mechanism, which effectively captures spectral features and spatial structures—critical for discriminating spectrally similar urban materials.

2) MUUFL Results: Our method achieved 86.37% OA, 86.12% AA, and 82.29% Kappa coefficient, improving upon the second-best GAMF by 1.11%, 6.14%, and 1.54%, respectively. The substantial AA improvement demonstrates excellent handling of class imbalance—a critical challenge in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON THE TRENTO DATASET (BEST RESULTS DISPLAYED IN BOLD)

NO.	GLT (2022)	HCT (2023)	ExViT (2023)	DSHF (2023)	GAMF (2024)	S2ENet (2024)	CrossHL (2024)	HLMamba (2024)	GFSFN (2025)	Ours
C1	99.51	98.64	99.27	**99.90**	99.12	99.62	99.66	98.93	99.60	**99.90**
C2	98.00	96.05	94.10	**99.93**	98.19	95.10	96.69	97.19	99.06	98.56
C3	98.38	99.04	98.25	99.78	99.56	98.30	97.22	97.82	**100.00**	97.94
C4	**100.00**	**100.00**	**100.00**	99.46	99.96	99.91	**100.00**	**100.00**	**100.00**	**100.00**
C5	**100.00**	99.15	**100.00**	99.99	99.81	**100.00**	99.86	99.98	99.95	**100.00**
C6	96.50	94.31	97.43	88.90	90.10	95.24	91.89	94.24	97.11	**98.08**
OA	99.34	98.53	99.04	98.65	98.59	98.93	98.70	98.94	99.49	**99.61**
AA	98.73	97.87	98.18	97.99	97.80	98.03	97.55	98.03	**99.23**	99.08
K	99.13	98.04	98.72	98.19	98.12	98.57	98.26	98.59	99.33	**99.48**

TABLE VII
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON THE AUGSBURG DATASET (BEST RESULTS DISPLAYED IN BOLD)

NO.	GLT (2022)	HCT (2023)	ExViT (2023)	DSHF (2023)	GAMF (2024)	S2ENet (2024)	CrossHL (2024)	HLMamba (2024)	GFSFN (2025)	Ours
C1	98.76	97.72	91.41	98.33	94.49	98.76	97.81	98.99	97.50	**99.47**
C2	85.28	87.21	86.67	84.62	86.41	91.85	79.93	84.38	85.96	**92.88**
C3	56.77	59.95	61.81	57.26	19.15	75.98	53.24	38.68	**68.20**	49.72
C4	86.61	89.59	78.58	85.41	88.98	77.24	82.23	88.24	88.20	**92.99**
C5	84.68	73.15	25.76	94.23	70.63	54.59	90.45	86.84	**95.31**	70.46
C6	55.93	63.26	53.90	48.8	81.84	41.47	50.27	52.49	61.16	**78.52**
C7	65.03	68.27	70.59	51.32	63.31	67.08	63.37	70.33	**75.82**	66.42
OA	85.66	87.53	82.07	84.8	84.74	86.07	81.64	85.07	87.21	**90.77**
AA	76.15	77.02	66.96	74.45	72.12	72.42	73.91	74.28	81.74	**78.47**
K	80.19	82.67	74.82	78.96	78.6	80.51	74.96	79.04	82.30	**86.95**

the MUUFL dataset, where the trees class contains 23 246 samples while yellow curb has only 183. HMMamba achieved 100% accuracy on water and 99.61% on cloth panels, while significantly improving minority classes such as yellow curb (67.51% versus GAMF's 52.74%). This balanced performance validates AWF's adaptive fusion strategy, which prevents majority class dominance through scene-specific modality weighting.

3) Trento Results: HMMamba achieved superior performance with 99.61% OA, 99.08% AA, and 99.48% Kappa coefficient, despite baseline methods approaching saturation. It surpassed GFSFN by 0.12% and 0.15% in OA and Kappa metrics, respectively. Perfect classification (100%) was achieved for woods and vineyard, with 99.90% for apple trees, demonstrating precise agricultural discrimination capability. Consistently high performance across all classes (minimum accuracy of 97.94% for ground) confirms that our multiscale approach effectively captures fine-grained spectral differences and structural patterns in agricultural landscapes.

4) Augsburg Results: HMMamba demonstrated the most significant improvements, achieving 90.77% OA, 78.47% AA, and 86.95% Kappa coefficient, exceeding the second-best HCT by 3.24%, 0.76%, and 4.28%, respectively. It attained the best accuracy in 6 out of 7 classes, particularly forest (99.47%) and residential area (92.88%), validating our method's effectiveness at coarse resolution (30 m). Although the industrial area class achieved 49.72% accuracy—lower than S2ENet's 75.98%—the substantial overall performance gain indicates that global context modeling effectively compensates for limited spatial detail. FFMB's linear complexity enables processing of larger spatial contexts, which is crucial for low-resolution classification.

5) Cross-Dataset Analysis: Performance patterns reveal method strengths across different scenarios. HMMamba ranked first in all OA and Kappa metrics across the four datasets, demonstrating robust generalization capability. The method particularly excels in: 1) complex urban environments (Houston2013 and Augsburg), where multiscale modeling and global dependencies are critical; 2) handling imbalanced datasets (MUUFL) by preventing bias through adaptive fusion; and 3) maintaining leadership in high-precision scenarios (Trento) through refined boundary delineation. Transformer-based methods (GLT, HCT, and CrossHL) and the recent GFSFN showed competitive results but gaps remain in comprehensive performance. CNN-based methods (DSHF) struggle with long-range dependencies. The graph-based GAMF performs well in structured scenes but lacks flexibility for diverse environments.

Consistent improvements across diverse datasets—from high-resolution urban (Houston2013) to low-resolution mixed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

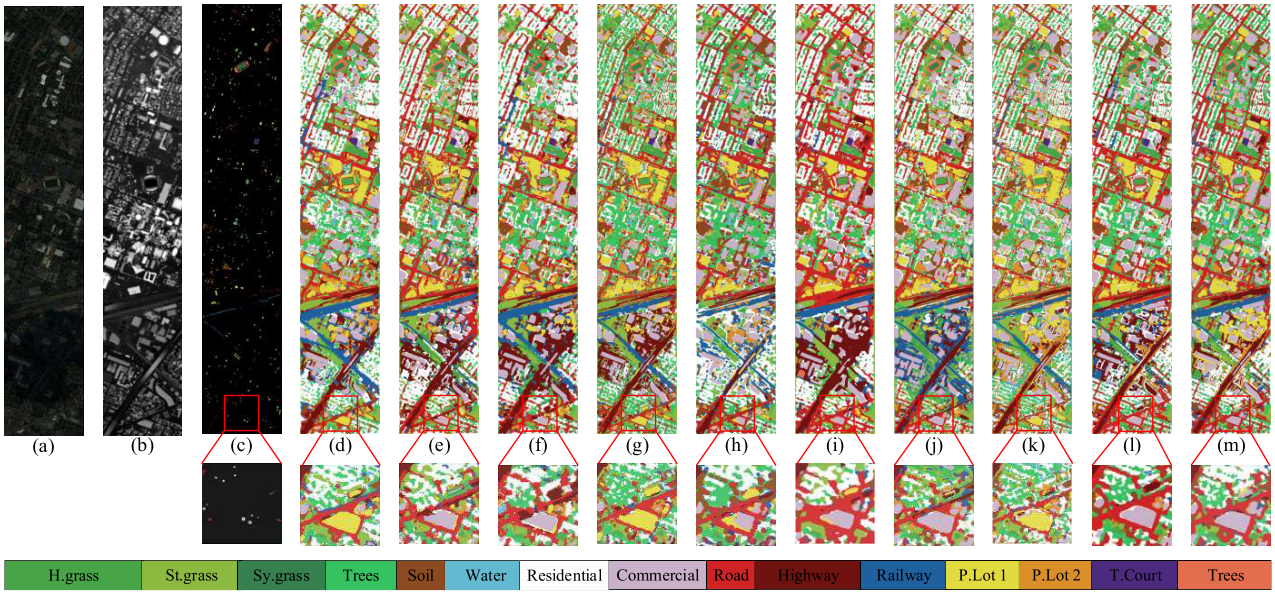


Fig. 7. Classification maps of all methods on the Houston2013 dataset. (a)–(m) Pseudo-color image for HSI, LiDAR-based DSM, ground truth, GLT, HCT, ExVit, DSHF, GAMF, S2ENet, CrossHL, HLMamba, GFSFN, and our method, respectively.

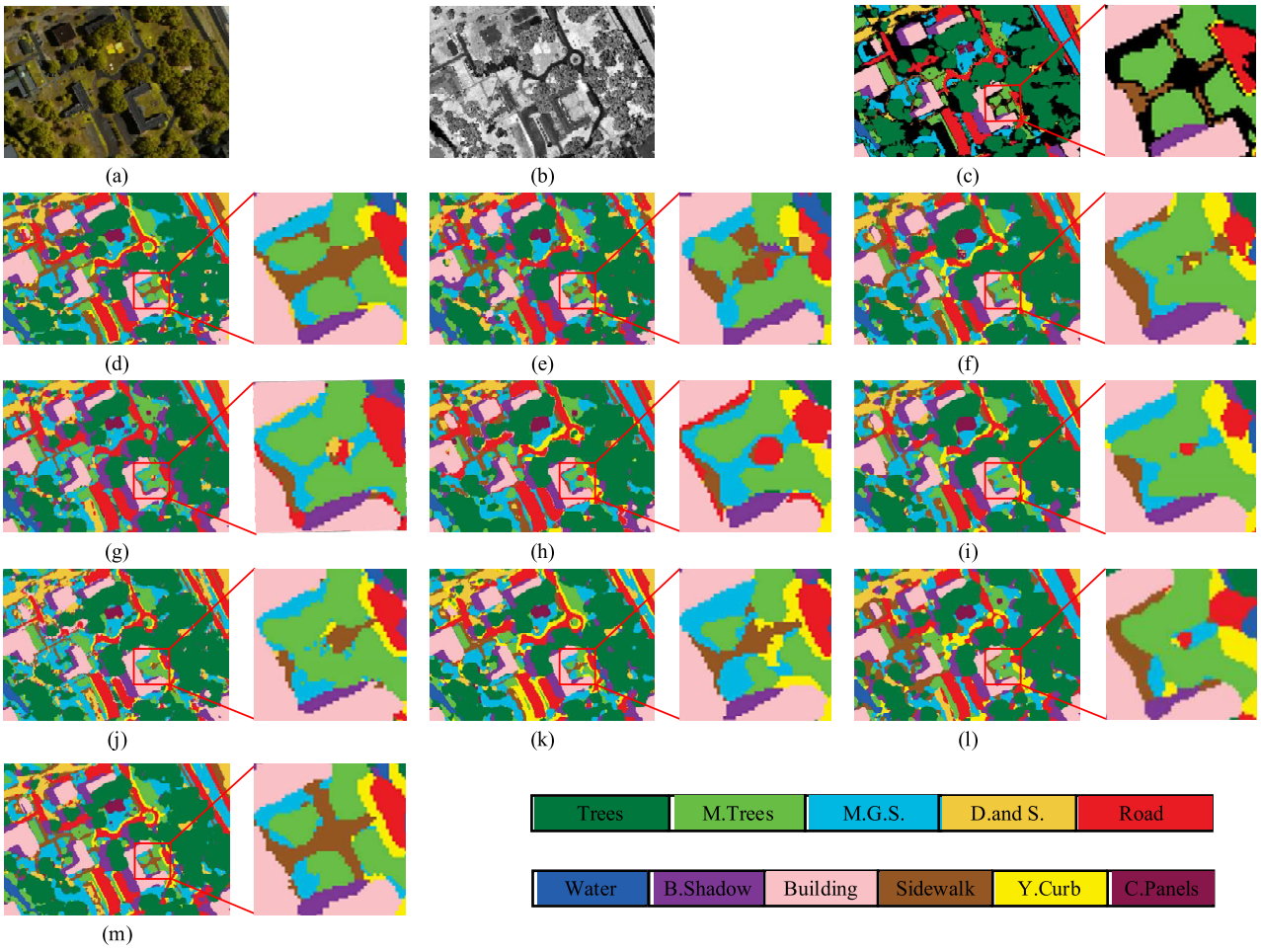


Fig. 8. Classification maps of all methods on the MUUFL dataset. (a)–(m) Pseudo-color image for HSI, LiDAR-based DSM, ground truth, GLT, HCT, ExVit, DSHF, GAMF, S2ENet, CrossHL, HLMamba, GFSFN, and our method, respectively.

(Augsburg), from balanced agricultural (Trento) to severely imbalanced campus (MUUFL)—validate HMMamba's architectural advantages: adaptive cross-modal fusion captures scene-specific modality importance, multiscale

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

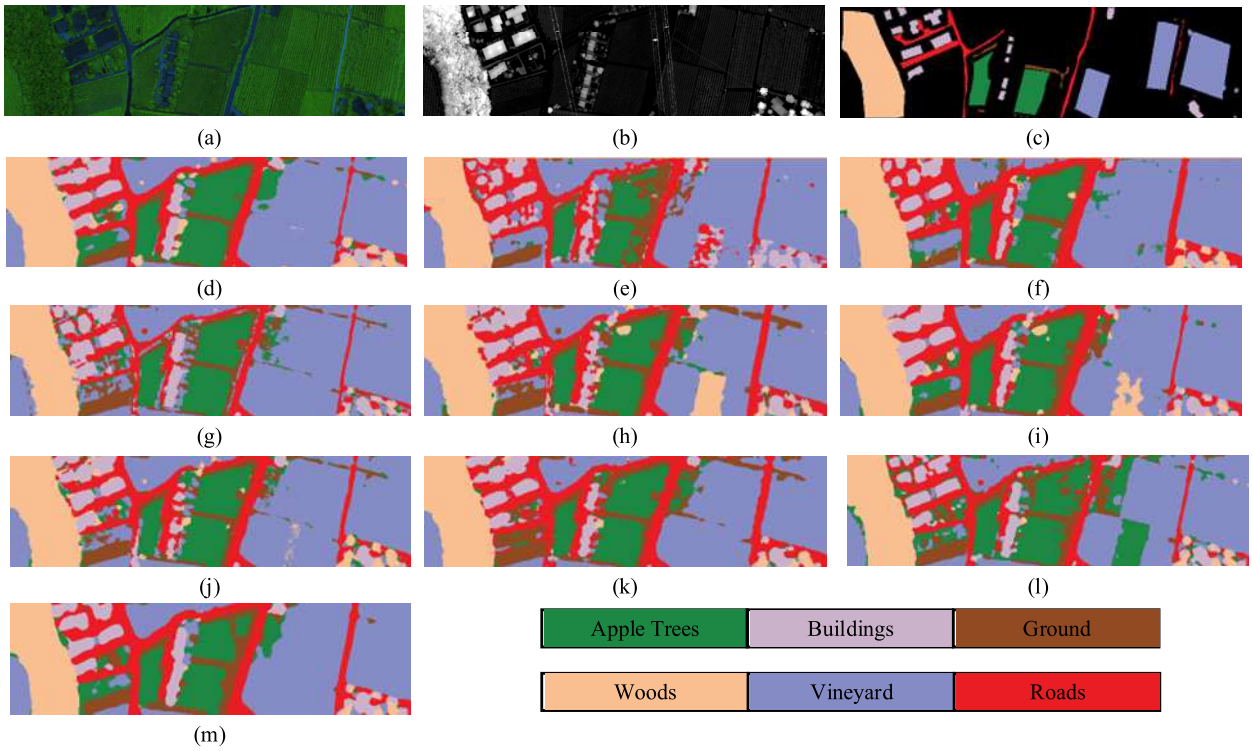


Fig. 9. Classification maps of all methods on the Trento dataset. (a)–(m) Pseudo-color image for HSI, LiDAR-based DSM, ground truth, GLT, HCT, ExVit, DSHF, GAMF, S2ENet, CrossHL, HLMamba, GFSFN, and our method, respectively.

attention selects optimal observation hierarchies, and Mamba's efficient global modeling achieves comprehensive spatial understanding without computational explosion.

6) Visualization Results and Analysis: Figs. 7–10 present the classification visualization results of all comparative methods across four datasets. These visualization regions are selected from representative typical scenes within the datasets. For instance, in the Houston2013 dataset, a segment covering the intersection of road networks, residential areas, and commercial districts is selected, while in the MUUFL dataset, an area containing buildings, shadows, and sidewalks is chosen. These selections facilitate a more intuitive demonstration of the model's performance under complex boundary conditions and mixed land-cover types.

a) Houston2013: As illustrated in Fig. 7, visual inspection reveals significant differences in urban structure preservation. GLT and HCT produce fragmented results with visible salt-and-pepper noise, particularly problematic for linear features such as roads and railways, creating unrealistic spatial discontinuities at class boundaries. ExViT and DSHF show improvement but struggle with small object discrimination—parking lots appear merged with surrounding roads, and tennis courts lose their rectangular geometry. GAMF and S2ENet better preserve object integrity but still exhibit edge roughness. CrossHL and HLMamba demonstrate cleaner boundaries, yet some confusion remains between spectrally similar classes (e.g., highway versus road). GFSFN shows competitive performance with improved spatial consistency, but classification artifacts persist in structurally complex regions.

HMMamba produces the most visually coherent map, maintaining road network connectivity, preserving building footprints, and clearly delineating recreational facilities. The significant improvements achieved in the commercial class (90.52% versus HLMamba's 87.49%) and road class (96.13% versus 93.98%) demonstrate superior boundary delineation capability in complex urban scenarios. These enhancements stem from FFMB's dual-branch attention mechanism, which effectively captures both spectral features and spatial structures crucial for discriminating spectrally similar urban materials.

b) MUUFL: As shown in Fig. 8, this dataset's primary challenge lies in distinguishing shadows from actual objects. Traditional methods frequently misclassify building shadows as separate structures or water bodies. HMMamba's classification map shows clear separation between buildings (red), building shadows (dark gray), and actual ground classes. The sidewalk network remains connected in our results while appearing fragmented in most comparison methods. This balanced performance validates AWF's adaptive fusion strategy, which prevents majority class dominance through scene-specific modality weighting.

c) Trento: Fig. 9 demonstrates the performance of various methods in agricultural scenarios. Despite high baseline accuracies, visual differences remain pronounced. Most methods produce blocky artifacts at agricultural parcel boundaries, failing to capture natural field variations. HMMamba generates smooth transitions between crop types while maintaining sharp boundaries where actual land use changes occur. Achieving perfect classification (100%) in woods and vineyard, along with 99.90% in apple trees, demonstrates precise agricultural

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

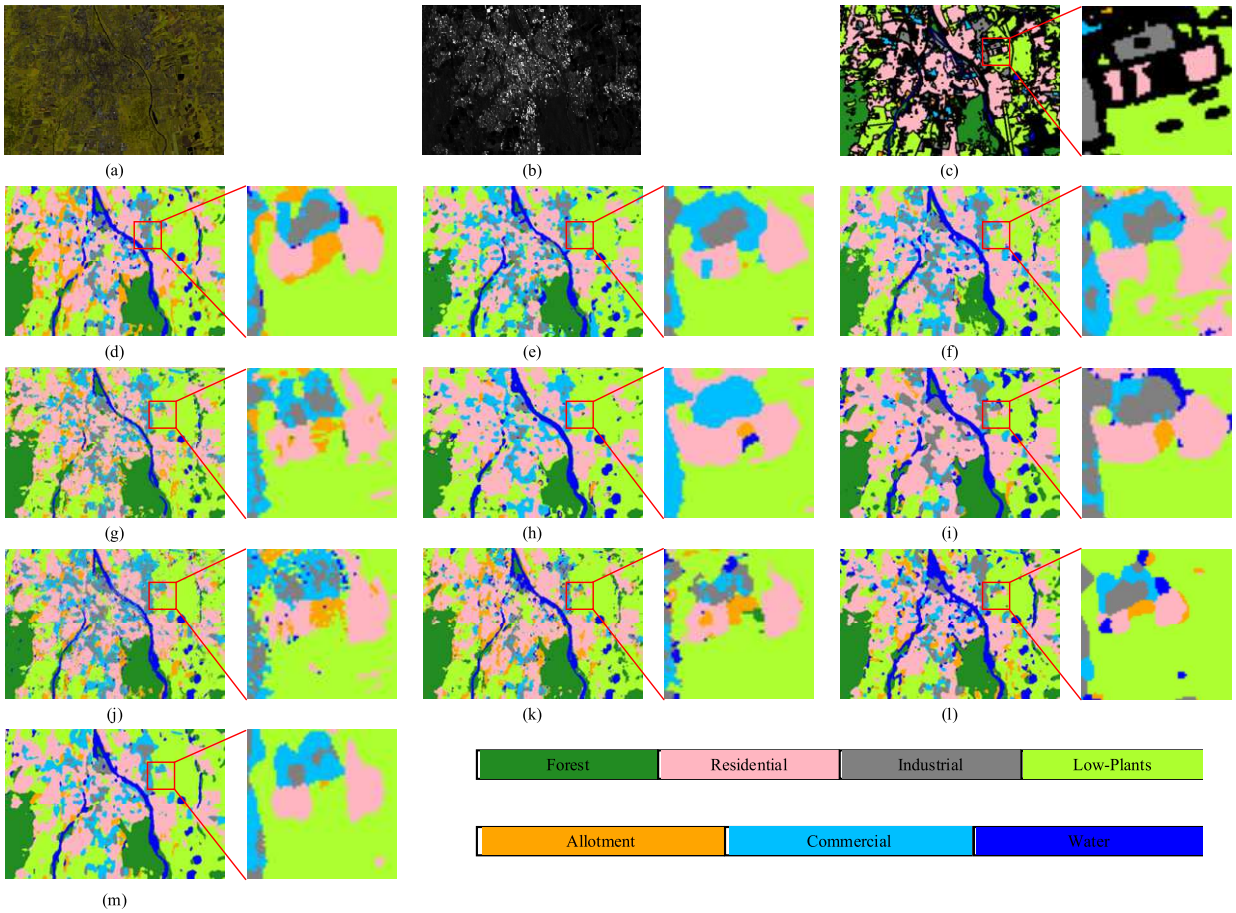


Fig. 10. Classification maps of all methods on the Augsburg dataset. (a)–(m) Pseudo-color image for HSI, LiDAR-based DSM, ground truth, GLT, HCT, ExVit, DSHF, GAMF, S2ENet, CrossHL, HLMamba, GFSFN, and our method, respectively.

discrimination capability. The vineyard rows' linear patterns are better preserved, and the distinction between woods and apple trees shows clearer demarcation without artificial pixelation.

d) Augsburg: As depicted in Fig. 10, low spatial resolution (30 m) amplifies classification challenges, making visual quality crucial for interpretation. Comparison methods show severe confusion at the forest-residential interface, with irregular boundaries suggesting classification uncertainty. Industrial areas appear fragmented in most results, incorrectly mixed with commercial zones. HMMamba produces notably cleaner maps with coherent urban structure: residential areas form connected neighborhoods, forests maintain natural boundaries, and water bodies appear as continuous regions rather than scattered pixels. The visual improvement directly correlates with our 3.02% OA gain, demonstrating that global context modeling effectively compensates for coarse resolution.

The consistent improvements across diverse datasets—from high-resolution urban (Houston2013) to low-resolution mixed landscapes (Augsburg), from balanced agricultural (Trento) to severely imbalanced campus (MUUFL)—validate HMMamba's architectural advantages: adaptive cross-modal fusion captures scene-specific modality importance, multiscale attention selects optimal observation hierarchies, and efficient global modeling through Mamba enables comprehensive spatial understanding without computational explosion.

7) T-Distributed Stochastic Neighbor Embedding (t-SNE) Feature Visualization: To intuitively analyze the feature representation capability, we employ t-SNE to visualize the features of four representative methods (see Figs. 11–14): GLT (transformer-based), DSHF (CNN-based), HLMamba (the latest Mamba-based method), and the proposed HMMamba. Each point represents a feature sample with color indicating its class label. Ideal feature distributions should exhibit tight intra-class clustering and clear interclass separation.

The visualization results reveal significant differences in feature quality. The GLT method shows relatively scattered feature distributions with low intra-class cohesion, reflecting its limitations in multimodal feature fusion. Although DSHF achieves certain interclass separation, obvious feature dispersion remains evident on the Houston2013 and Augsburg datasets, indicating limited discriminative capability. HLMamba achieves improved visualization results with most classes forming relatively compact clusters.

In contrast, HMMamba demonstrates optimal feature distribution patterns across all datasets. On the Trento dataset (see Fig. 13), the six classes form significantly clearer boundaries and compact cluster structures with minimal interclass overlap. On the MUUFL dataset (see Fig. 12), despite severe

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

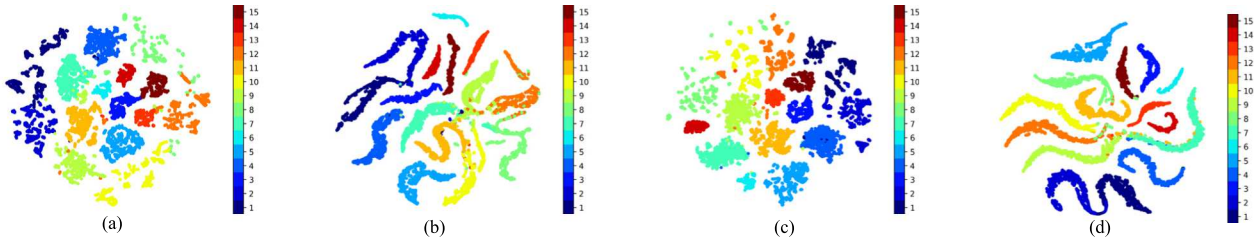


Fig. 11. Feature visualization via t-SNE on the Houston2013 dataset. (a) GLT, (b) DSHF, (c) HLMamba, and (d) our method.

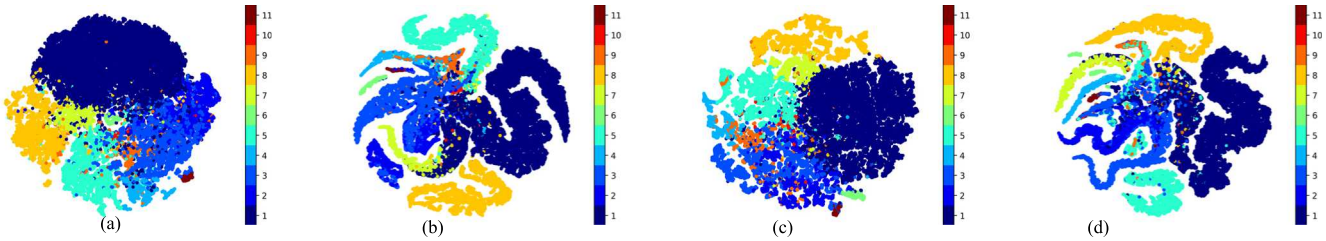


Fig. 12. Feature visualization via t-SNE on the MUUFL dataset. (a) GLT, (b) DSHF, (c) HLMamba, and (d) our method.

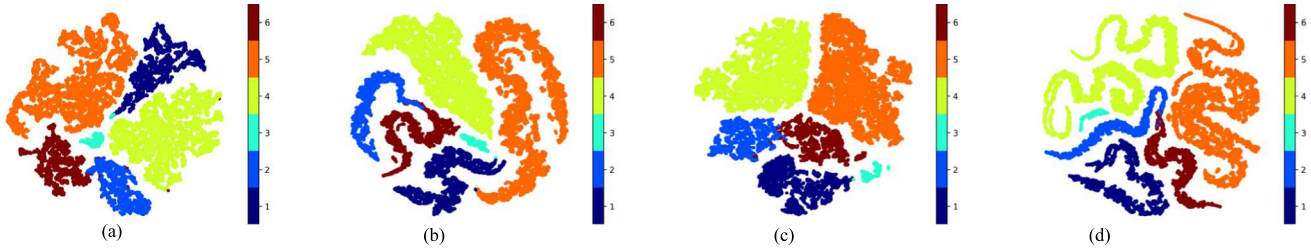


Fig. 13. Feature visualization via t-SNE on the Trento dataset. (a) GLT, (b) DSHF, (c) HLHMamba, and (d) our method.

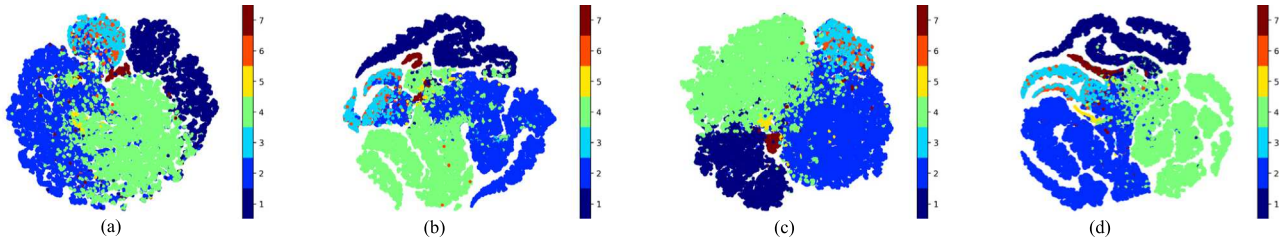


Fig. 14. Feature visualization via t-SNE on the Augsburg dataset. (a) GLT, (b) DSHF, (c) HLMamba, and (d) our method.

class imbalance, the clustering degree of each class remains significantly superior to other methods. The visualizations for Houston2013 (see Fig. 11) and Augsburg (see Fig. 14) datasets also exhibit more compact intra-class distributions and larger interclass margins.

These visualization results validate that HMMamba effectively integrates complementary HSI-LiDAR information through a hierarchical multiscale fusion architecture and enhances feature discriminability through the FFMB module, ultimately generating feature representations with high intra-class compactness and interclass separability, providing a solid foundation for high-precision classification.

V. Conclusion

This article proposes an HMMamba network to address critical challenges in HSI-LiDAR joint classification: cross-modal adaptive fusion, multiscale collaboration, and efficient global modeling. Three key modules are introduced: AWF achieves scene-adaptive cross-modal fusion through dual-level weight control; MSAF automatically selects optimal observation scales via competitive attention mechanisms; FFMB combines dual-branch attention with selective state space modeling to capture global dependencies at $O(N)$ linear complexity.

Experiments on four benchmark datasets demonstrate that HMMamba achieves optimal performance across all datasets, with OA improvements of 0.12%–3.24%, AA improvements of 0.42%–6.05%, and Kappa coefficient improvements of 0.15%–6.44%. The method excels particularly in complex urban environments and class-imbalanced scenarios. Ablation studies confirm module effectiveness, achieving significant performance gains with only 1.8% computational overhead. Compared to transformer-based methods, HMMamba reduces complexity from $O(N^2)$ to $O(N)$; compared to CNN-based methods, it significantly enhances long-range dependency capture.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.

Current limitations include improvable discrimination of highly spectrally similar classes, and constrained multiscale strategy effectiveness under extremely low spatial resolution scenarios. Future work will explore deeper cross-modal interaction mechanisms for fine-grained discrimination, investigate model compression techniques for deployment efficiency, and extend to multitemporal remote sensing classification and change detection tasks.

REFERENCES

[1] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020, doi: 10.1016/j.isprsjprs.2020.06.014.

[2] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[3] J. M. Bioucas-Dias et al., "Hyperspectral remote sensing data analysis and future challenges," *Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, 2013.

[4] P. Ghamisi et al., "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3028, 2017.

[5] Z. Wang and M. Menenti, "Challenges and opportunities in LiDAR remote sensing," *Frontiers Remote Sens.*, vol. 2, Mar. 2021, Art. no. 641723.

[6] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, 2014.

[7] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, 2016.

[8] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Jun. 2019.

[9] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, 2017.

[10] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[11] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[12] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[13] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[14] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[15] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[16] S. K. Roy, A. Sukul, A. Jamali, J. M. Haut, and P. Ghamisi, "Cross hyperspectral and LiDAR attention transformer: An extended self-attention for land use and land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[17] Y. Feng, J. Zhu, R. Song, and X. Wang, "S2EFT: Spectral–spatial-elevation fusion transformer for hyperspectral image and LiDAR classification," *Knowl.-Based Syst.*, vol. 283, Jan. 2024, Art. no. 111190.

[18] F. Zhu, C. Shi, K. Shi, and L. Wang, "Joint classification of hyperspectral and LiDAR data using hierarchical multimodal feature aggregation-based multihead axial attention transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[19] H. Wu, Z. Du, D. Zhong, Y. Wang, and C. Tao, "FSVLM: A vision-language model for remote sensing farmland segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[20] H. Wu, W. Mu, D. Zhong, Z. Du, H. Li, and C. Tao, "FarmSeg_VLM: A farmland remote sensing image segmentation method considering vision-language alignment," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 423–439, Jul. 2025.

[21] J. Cai et al., "A novel graph-attention based multimodal fusion network for joint classification of hyperspectral image and LiDAR data," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123587.

[22] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[23] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[24] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[25] X. Ma, X. Zhang, and M. O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 6011405.

[26] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multiscale feature fusion state space model for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504116, doi: 10.1109/TGRS.2025.3535622.

[27] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data based on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530915, doi: 10.1109/TGRS.2024.3459709.

[28] C. Shi, F. Zhu, K. Shi, L. Wang, and H. Pan, "TBi-mamba: Rethinking joint classification of hyperspectral and LiDAR data with bidirectional mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5522515.

[29] J. Lin, X. Jin, F. Gao, J. Dong, and H. Yu, "Boosting spatial–spectral masked auto-encoder through mining redundant spectra for HSI-SAR/LiDAR classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Athens, Greece, Jul. 2024, pp. 9744–9747, doi: 10.1109/IGARSS53475.2024.10642405.

[30] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816, doi: 10.1109/TGRS.2023.3236154.

[31] F. Gao et al., "Prototype-based information compensation network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513615, doi: 10.1109/TGRS.2025.3569335.

[32] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[33] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[34] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.

[35] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[36] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, Jul. 2020.

[37] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, Feb. 2018.

[38] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, Jul. 2021.

[39] C. Shi, Z. Zhong, S. Ding, Y. Lei, L. Wang, and Z. Jin, "Multilevel feature gated fusion based spatial and frequency domain attention network for joint classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5960–5974, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:25:26 UTC from IEEE Xplore. Restrictions apply.